Lesson 37: Learning in Games and Bayesian Games

AI/ML Mathematics

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1 Motivation and Overview

The Nash equilibrium concept tells us what stable play *looks like*, but says nothing about how players might *arrive* at it. This lesson develops two classical learning dynamics—fictitious play and multiplicative weights update—and establishes their convergence in zero-sum games, then introduces Bayesian games modeling strategic interaction under incomplete information.

Both parts share a common theme: *uncertainty about the opponent.* In learning dynamics the uncertainty concerns the opponent's strategy; in Bayesian games it concerns the opponent's type.

From Game Learning to Multi-Agent RL

Fictitious play is the prototype for self-play algorithms (NFSP, PSRO), while MWU is a precursor to policy gradient methods in competitive settings. The Bayesian game framework underpins opponent modeling, where an agent maintains beliefs about opponent types and updates via Bayes' rule. These classical results provide the theoretical foundation for modern MARL algorithms.

2 References

- **Fictitious play:** Robinson [1951] proves convergence in zero-sum games; Fudenberg and Tirole [1991], Chapter 2.
- **No-regret learning:** Freund and Schapire [1999] develop MWU for game playing; Shoham and Leyton-Brown [2008], Chapter 7.
- **Bayesian games:** Harsanyi [1967] introduces the type-based formulation; Osborne and Rubinstein [1994], Chapter 9.
- **Nash existence:** Nash [1950, 1951].

3 Fictitious Play

Definition 3.1 (Fictitious Play)**.** Consider a finite two-player game $G = (\{1,2\}, A_1, A_2, g_1, g_2)$. *Fictitious play* (FP) is a learning dynamic where each player best-responds to the empirical frequency of the opponent's past actions. The empirical frequency of player j after T rounds is

$$\hat{\alpha}_j^T(a_j) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{a_j^t = a_j\}, \qquad a_j \in A_j.$$

At round $T+1$, player i plays $a_i^{T+1} \in \mathrm{BR}_i(\hat{\alpha}_j^T)$.

Remark 3.2. Fictitious play requires no knowledge of the opponent's payoff function—only observation of the opponent's past actions. Each player acts *as if* the opponent plays a fixed mixed strategy equal to the empirical frequency, and best-responds to this belief. The "fictitious" qualifier refers to this assumption: the opponent is not actually stationary, but is simultaneously updating.

We work with two-player zero-sum games where $g_1 = g = -g_2$ and $v^* = \max_{\alpha_1} \min_{\alpha_2} \mathbb{E}[g(\alpha_1, \alpha_2)]$ is the value (Lesson 34, minimax theorem).

Theorem 3.3 (FP Convergence in 2×2 Zero-Sum Games)**.** *In any 2×2 zero-sum game, the empirical frequencies under fictitious play converge to a Nash equilibrium as $T \to \infty$.*

Proof. Let $R_k^T = \sum_{t=1}^T g(k, a_2^t)$ for rows $k \in \{1, 2\}$ and $D^T = R_1^T - R_2^T$. Player 1 plays row 1 when $D^T > 0$ and row 2 when $D^T < 0$. Player 2's best response to the heavier-weighted row tends to decrease $|D^T|$. Each step changes D^T by at most $C_0 = \max_{a_2} |g(1, a_2) - g(2, a_2)|$, and the dynamics push D back toward zero, so $|D^T|$ remains bounded.

Since $|D^T|$ is bounded, $D^T/T \to 0$. Define $V_1^T = \frac{1}{T} \max_{a_1} \sum_{t=1}^T g(a_1, a_2^t) = \frac{1}{T} \max\{R_1^T, R_2^T\}$. We have $V_1^T \geq v^*$ and $V_1^T = (R_1^T + R_2^T)/(2T) + |D^T|/(2T) \to \bar{r}$. The analogous quantity V_2^T for player 2 satisfies $V_1^T + V_2^T \to 0$, giving $V_1^T \to v^*$.

This forces $\hat{\alpha}_2^T$ to converge to a distribution making player 1 indifferent (the NE condition), and symmetrically for $\hat{\alpha}_1^T$. □

Robinson's Theorem

Theorem 3.4 (Robinson's Theorem)**.** *In any finite two-player zero-sum game, the empirical frequencies under fictitious play converge to the set of Nash equilibria.*

Proof. Define $\Phi_1^T = \max_{a_1} \sum_{t=1}^T g(a_1, a_2^t)$ and $\Phi_2^T = \max_{a_2} \sum_{t=1}^T (-g(a_1^t, a_2))$. Then $\Phi_1^T/T \geq v^*$ and $\Phi_2^T/T \geq -v^*$.

Step 1. We show $(\Phi_1^T + \Phi_2^T)/T \to 0$. Let $S^T = \sum_{t=1}^T g(a_1^t, a_2^t)$. Then $\Phi_1^T \geq S^T$ and $\Phi_2^T \geq -S^T$, so $\Psi^T := \Phi_1^T + \Phi_2^T \geq 0$. For the upper bound, the best-response property gives $g(a_1^{T+1}, \hat{\alpha}_2^T) \geq \Phi_1^T/T$ and $-g(\hat{\alpha}_1^T, a_2^{T+1}) \geq \Phi_2^T/T$. Tracking the increments $\Psi^{T+1} - \Psi^T$ carefully (Robinson's original argument), one shows $\Psi^T = O(\sqrt{T})$, hence $\Psi^T/T \to 0$.

Step 2. Since $\Phi_1^T/T \geq v^*$, $\Phi_2^T/T \geq -v^*$, and their sum divided by T tends to zero: $\Phi_1^T/T \to v^*$ and $\Phi_2^T/T \to -v^*$. Any limit point (α_1^*, α_2^*) of the empirical frequencies (which exists by compactness) satisfies $\max_{a_1} g(a_1, \alpha_2^*) = v^* = \min_{a_2} g(\alpha_1^*, a_2)$, the NE condition. □

Fictitious Play and PSRO

Fictitious play is the ancestor of *Policy-Space Response Oracles* (PSRO) in modern MARL: each player maintains a population of policies and computes a best response to the meta-mixture, exactly as in FP but with neural-network policies. Robinson's theorem guarantees convergence to NE in zero-sum games.

4 No-Regret Learning and Multiplicative Weights

Definition 4.1 (External Regret)**.** Player i's *external regret* after T rounds is

$$R_i^T = \max_{a_i \in A_i} \sum_{t=1}^T g_i(a_i, a_{-i}^t) - \sum_{t=1}^T g_i(\alpha_i^t, a_{-i}^t),$$

comparing cumulative payoff to the best fixed action in hindsight. A learning algorithm is *no-regret* (Hannan consistent) if $R_i^T/T \to 0$ regardless of the opponent's sequence.

Definition 4.2 (Multiplicative Weights Update)**.** Let $K = |A_i|$ and $\eta > 0$ be a learning rate. The *multiplicative weights update* (MWU) maintains weights $w^t(a_i)$ for each action $a_i \in A_i$:

$$w^1(a_i) = 1 \quad \text{for all } a_i \in A_i,$$
$$w^{t+1}(a_i) = w^t(a_i) \cdot \exp\big(\eta \cdot g_i(a_i, a_{-i}^t)\big).$$

The mixed strategy at time t is $\alpha_i^t(a_i) = w^t(a_i)/W^t$, where $W^t = \sum_{a_i' \in A_i} w^t(a_i')$.

Remark 4.3. MWU increases the weight of actions that performed well and decreases the weight of those that performed poorly. The exponential update ensures weights remain positive. When η is small, updates are conservative; when η is large, the algorithm tracks the best action aggressively but may overshoot.

MWU Regret Bound

Theorem 4.4 (MWU Regret Bound)**.** *If $|g_i(a)| \le 1$ for all a, then with $\eta = \sqrt{(\ln K)/T}$, MWU achieves $R_i^T \le 2\sqrt{T \ln K}$.*

Proof. We use a potential function argument with $W^t = \sum_{a_i} w^t(a_i)$, so $W^1 = K$.

Upper bound on $\ln(W^{T+1}/W^1)$**.**

$$\frac{W^{t+1}}{W^t} = \sum_{a_i} \alpha_i^t(a_i) \exp\big(\eta\, g_i(a_i, a_{-i}^t)\big).$$

Since $|g_i| \le 1$ and $\eta \le 1$ (for $T \ge \ln K$), we have $|\eta g_i| \le 1$. Using $e^x \le 1 + x + x^2$ for $|x| \le 1$:

$$\frac{W^{t+1}}{W^t} \le 1 + \eta\, g_i(\alpha_i^t, a_{-i}^t) + \eta^2.$$

Since $1 + x \le e^x$, taking logs and telescoping over $t = 1, \dots, T$:

$$\ln \frac{W^{T+1}}{W^1} \le \eta \sum_{t=1}^{T} g_i(\alpha_i^t, a_{-i}^t) + \eta^2 T.$$

Lower bound on $\ln(W^{T+1}/W^1)$**.** For any fixed action $a_i^* \in A_i$: $W^{T+1} \ge w^{T+1}(a_i^*) = \exp\big(\eta \sum_{t=1}^T g_i(a_i^*, a_{-i}^t)\big)$, so $\ln(W^{T+1}/W^1) \ge \eta \sum_t g_i(a_i^*, a_{-i}^t) - \ln K$.

Combining. Rearranging: $\sum_t g_i(a_i^*, a_{-i}^t) - \sum_t g_i(\alpha_i^t, a_{-i}^t) \le (\ln K)/\eta + \eta T$. Taking the max over a_i^*: $R_i^T \le (\ln K)/\eta + \eta T$. Substituting $\eta = \sqrt{(\ln K)/T}$: $R_i^T \le 2\sqrt{T \ln K}$. □

MWU and Policy Gradient

The MWU update $\alpha_i^{t+1} \propto \alpha_i^t \exp(\eta\, g_i(\cdot, a_{-i}^t))$ is precisely the REINFORCE policy gradient update with softmax parameterization. The $O(\sqrt{T})$ regret bound provides convergence rate guarantees for policy gradient in competitive games.

5 No-Regret Learning Implies NE in Zero-Sum Games

No-Regret to Nash Equilibrium

Theorem 5.1 (No-Regret Implies NE in Zero-Sum Games)**.** *In a two-player zero-sum game, if both players use no-regret algorithms, the time-averaged strategies $\bar{\alpha}_i^T = \frac{1}{T}\sum_t \alpha_i^t$ converge to the set of Nash equilibria.*

Proof. By no-regret for player 1 and linearity: $\max_{a_1} g(a_1, \bar{\alpha}_2^T) \le \frac{1}{T}\sum_t g(\alpha_1^t, \alpha_2^t) + \varepsilon_1^T$ where $\varepsilon_1^T = R_1^T/T \to 0$. Similarly for player 2: $\min_{a_2} g(\bar{\alpha}_1^T, a_2) \ge \frac{1}{T}\sum_t g(\alpha_1^t, \alpha_2^t) - \varepsilon_2^T$.

Combining: $\min_{a_2} g(\bar{\alpha}_1^T, a_2) \ge \max_{a_1} g(a_1, \bar{\alpha}_2^T) - \varepsilon_1^T - \varepsilon_2^T$. The minimax inequality gives the reverse direction, so the duality gap $\max_{a_1} g(a_1, \bar{\alpha}_2^T) - \min_{a_2} g(\bar{\alpha}_1^T, a_2) \le \varepsilon_1^T + \varepsilon_2^T \to 0$. Any limit point satisfies the NE condition. □

Corollary 5.2. *With both players using MWU, the duality gap is at most $4\sqrt{(\ln K)/T}$, giving $O(1/\sqrt{T})$ convergence to NE.*

6 Bayesian Games

Definition 6.1 (Bayesian Game)**.** A *Bayesian game* (or game of incomplete information) is a tuple $\Gamma = (N, (A_i)_{i\in N}, (\Theta_i)_{i\in N}, p, (u_i)_{i\in N})$ where:

(a) $N = \{1, \ldots, n\}$ is the player set.

(b) A_i is the finite action set of player i.

(c) Θ_i is the finite *type space* of player i.

(d) $p \in \Delta(\Theta)$ is the *common prior* over the type profile $\theta = (\theta_1, \ldots, \theta_n) \in \Theta = \prod_i \Theta_i$.

(e) $u_i : A \times \Theta \to \mathbb{R}$ is player i's payoff.

The timing is: nature draws $\theta \sim p$; each player i observes only θ_i; players simultaneously choose actions.

Remark 6.2. The *common prior assumption* means all players agree on p. Player i's posterior belief is $p(\theta_{-i}|\theta_i) = p(\theta_i, \theta_{-i}) / \sum_{\theta'_{-i}} p(\theta_i, \theta'_{-i})$. Harsanyi [1967] showed any incomplete-information game can be modeled this way.

Definition 6.3 (Bayesian Nash Equilibrium)**.** A *(pure) strategy* in a Bayesian game is a function $s_i : \Theta_i \to A_i$ mapping types to actions. A strategy profile $s^* = (s_1^*, \ldots, s_n^*)$ is a *Bayesian Nash equilibrium* (BNE) if, for every player i and every type $\theta_i \in \Theta_i$ with $p(\theta_i) > 0$:

$$s_i^*(\theta_i) \in \arg\max_{a_i \in A_i} \mathbb{E}_{\theta_{-i}\sim p(\cdot|\theta_i)} \big[u_i(a_i, s_{-i}^*(\theta_{-i}), \theta_i, \theta_{-i})\big].$$

A *mixed* BNE allows $s_i(\theta_i) \in \Delta(A_i)$.

Remark 6.4. The BNE condition says: each type of each player best-responds to the *expected* play of opponents, where the expectation is over opponents' types (determining their actions through s_{-i}^*) using the posterior $p(\theta_{-i}|\theta_i)$.

BNE Existence

Theorem 6.5 (Existence of BNE)**.** *Every finite Bayesian game has a BNE in mixed strategies.*

Proof. For each pair (i, θ_i), create an "agent" choosing from A_i with payoff $\tilde{u}_{i,\theta_i} = \sum_{\theta_{-i}} p(\theta_{-i}|\theta_i)\, u_i(a_{i,\theta_i}, a_{-i,\theta_{-i}}, \theta$ This *agent normal form* is a finite game, so Nash's theorem (Lesson 34) guarantees a mixed NE. Translating back: $s_i^*(\theta_i) = \sigma_{i,\theta_i}^*$ is a BNE, since the NE condition for agent (i, θ_i) is exactly the BNE condition. □

7 Examples

Example 7.1 (Cournot Duopoly with Unknown Costs)**.** Inverse demand $P(Q) = a - q_1 - q_2$. Firm 1 has known cost c_1; firm 2 has private cost $c_2 \in \{c_L, c_H\}$ with $\mathbb{P}(c_2 = c_L) = \pi$. Profit: $u_i = (a - q_1 - q_2 - c_i)q_i$. The BNE first-order conditions are:

$$q_1 = \tfrac{1}{2}[a - c_1 - \pi\, s_2(c_L) - (1-\pi)s_2(c_H)], \tag{1}$$
$$s_2(c) = \tfrac{1}{2}(a - c - q_1) \quad \text{for } c \in \{c_L, c_H\}. \tag{2}$$

Substituting (2) into (1) with $\bar{c} = \pi c_L + (1-\pi)c_H$ gives $q_1 = (a - 2c_1 + \bar{c})/3$. Firm 1's quantity depends on the *expected* cost; firm 2 conditions on its actual type.

Example 7.2 (First-Price Sealed-Bid Auction)**.** Two bidders with $v_i \sim \text{Uniform}[0,1]$ independently. Highest bidder wins and pays her bid: $u_i = (v_i - b_i)\mathbf{1}\{b_i > b_j\}$. Seeking symmetric BNE $b_i = \beta(v_i)$ with β increasing, the FOC yields the ODE $(v\beta)' = v$ with $\beta(0) = 0$, giving $\beta(v) = v/2$: each bidder bids half her value.

Verification. Against $\beta(v_j) = v_j/2$, bidder i solves $\max_b(v_i - b) \cdot 2b$. The FOC gives $2v_i - 4b = 0$, hence $b = v_i/2$, confirming the BNE.

The equilibrium bidding function $\beta(v) = v/2$ shows that each bidder "shades" her bid below her true value. The extent of shading depends on the number of competitors and the value distribution.

Opponent Modeling as Bayesian Inference

The Bayesian framework formalizes opponent modeling in MARL: type θ_j encodes the opponent's policy parameters; the prior $p(\theta_{-i})$ represents initial beliefs; Bayes' rule updates beliefs as actions are observed; the BNE condition requires maximizing expected return under the current posterior. Algorithms like LOLA and Bayesian policy reuse implement this framework.

8 Exercises

Exercise 8.1 (Fictitious Play in Matching Pennies (*))**.** Consider the zero-sum game Matching Pennies:

	H	T
H	$1, -1$	$-1, 1$
T	$-1, 1$	$1, -1$

(a) Compute the unique Nash equilibrium and the value of the game.

(b) Simulate fictitious play for 10 rounds starting with both players choosing H. Record the empirical frequencies at each step.

(c) Verify that the empirical frequencies approach the NE.

Exercise 8.2 (MWU Weight Computation (**))**.** Player 1 has $A_1 = \{A, B, C\}$. The payoff sequence for rounds $t = 1, 2, 3$ is $(1, 0, -1)$, $(-1, 1, 0)$, $(0, -1, 1)$.

(a) Starting with $w^1 = (1, 1, 1)$ and $\eta = 0.5$, compute weights and mixed strategies for $t = 1, 2, 3, 4$.

(b) Compute the external regret R^3.

(c) Compare with the theoretical bound $2\sqrt{3 \ln 3}$.

Exercise 8.3 (No-Regret in General-Sum Games (**))**.** Consider the Prisoner's Dilemma:

	L	R
T	$3, 3$	$0, 4$
B	$4, 0$	$1, 1$

(a) Find all Nash equilibria of this game.

(b) Suppose both players use no-regret learning. Show that the time-averaged play can converge to a correlated distribution placing high weight on (T, L), which is not a NE.

(c) Explain why Theorem 5.1 fails for general-sum games.

Exercise 8.4 (BNE Computation (**)). Player 1 has type $\theta_1 \in \{H, L\}$ with $\mathbb{P}(H) = 1/2$. Player 2 has no private information. Actions: $A_1 = \{U, D\}$, $A_2 = \{L, R\}$. Payoffs:

Type H:

	L	R
U	$2,1$	$0,0$
D	$0,0$	$1,2$

Type L:

	L	R
U	$0,0$	$1,2$
D	$2,1$	$0,0$

(a) Formulate this as a Bayesian game and identify the agent normal form.

(b) Find all pure-strategy BNE.

(c) Find a mixed-strategy BNE where player 2 randomizes.

Exercise 8.5 (Second-Price Auction (*)). In a second-price sealed-bid auction with two bidders having private values $v_i \sim \text{Uniform}[0, 1]$, the highest bidder wins and pays the second-highest bid.

(a) Show that bidding $b_i = v_i$ (truthful bidding) is a weakly dominant strategy for each bidder.

(b) Conclude that truthful bidding is a BNE.

(c) Compare the expected seller revenue with the first-price auction of Example 7.2.

Exercise 8.6 (MWU Convergence Rate (**)). In Rock-Paper-Scissors (payoffs $0, \pm 1$):

(a) Compute the unique NE and the value of the game.

(b) If both players use MWU with $\eta = \sqrt{(\ln 3)/T}$, bound the duality gap after $T = 1000$ rounds.

(c) How many rounds T are needed for the duality gap to drop below 0.01?

Exercise 8.7 (Cournot Duopoly BNE (***)). In Example 7.1, let $a = 10$, $c_1 = 2$, $c_L = 1$, $c_H = 3$, and $\pi = 1/2$.

(a) Solve the system (1)–(2) for q_1, $s_2(c_L)$, and $s_2(c_H)$.

(b) Compute each firm's expected profit in the BNE.

(c) Compare with the complete-information outcomes: (i) $c_2 = c_L$ known, (ii) $c_2 = c_H$ known.

(d) Does incomplete information help or hurt firm 1? Explain.

Exercise 8.8 (FP Non-Convergence in General-Sum Games (***)). Shapley (1964) showed FP can fail to converge. Consider:

	L	C	R
T	$0,0$	$1,0$	$0,1$
M	$0,1$	$0,0$	$1,0$
B	$1,0$	$0,1$	$0,0$

(a) Verify the unique NE is $(1/3, 1/3, 1/3)$ for each player.

(b) Simulate FP for 20 rounds from (T, L).

(c) Show the empirical frequencies cycle without converging.

(d) Discuss implications for using FP in general MARL settings.

9 Summary

- **Fictitious play:** best-respond to empirical frequencies. Converges to NE in 2×2 zero-sum games (Theorem 3.3) and all finite zero-sum games (Robinson, Theorem 3.4).
- **MWU** achieves regret $R^T \leq 2\sqrt{T \ln K}$ (Theorem 4.4).
- **No-regret implies NE** in zero-sum games at rate $O(1/\sqrt{T})$ (Theorem 5.1).
- **Bayesian games** model incomplete information via types; **BNE** requires each type to best-respond given posterior beliefs. Existence in finite games via agent normal form (Theorem 6.5).
- Examples: Cournot duopoly with unknown costs, first-price auctions with bid shading $\beta(v) = v/2$.

10 Further Reading

- **Lesson 34**: Nash existence and the minimax theorem.
- **Lesson 36**: Repeated games and folk theorems.
- **Lesson 38**: Signaling, PBE, and potential games.
- Fudenberg and Tirole [1991], Ch. 2 (learning) and Ch. 6 (Bayesian).
- Freund and Schapire [1999]: MWU and connections to boosting.
- Shoham and Leyton-Brown [2008], Ch. 7–8: learning in multi-agent systems.

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